

Experiment No.	5
Experiment Title	Decision Tree and Random Forest: A Comparative Classification Study ¹
Student Name	Pranav N
Register Number	3122247001044
Faculty	Dr. Poreddy Ajay Kumar Reddy
Submission Date	Aug 2026

1 Objective

- To implement a Decision Tree classifier.
- To extend the Decision Tree into a Random Forest ensemble model.
- To study the impact of hyperparameters on overfitting and generalization.
- To select optimal hyperparameters using 5-Fold Cross-Validation.
- To compare single-tree and ensemble-tree models.

2 Problem Statement

Given 30 numerical features describing cell nuclei extracted from digitized images of breast masses, the task is to classify each sample as Malignant or Benign. A single Decision Tree classifier and a Random Forest ensemble classifier are trained, hyperparameters are tuned using 5-fold cross-validation, and the two models are compared using standard classification metrics to study how ensembling affects overfitting and generalization.

3 Dataset Description

Dataset Name	Wisconsin Diagnostic Breast Cancer (WDBC) Dataset
Dataset Source	scikit-learn <code>load_breast_cancer()</code> (UCI ML Repository)
Number of Samples	569
Number of Features	30 numerical attributes
Number of Classes	2 (Malignant = 0, Benign = 1)
Missing Values	None (0 missing values found)
Train-Test Split	80–20 (455 training samples, 114 testing samples), stratified

4 Exploratory Data Analysis

Class Distribution

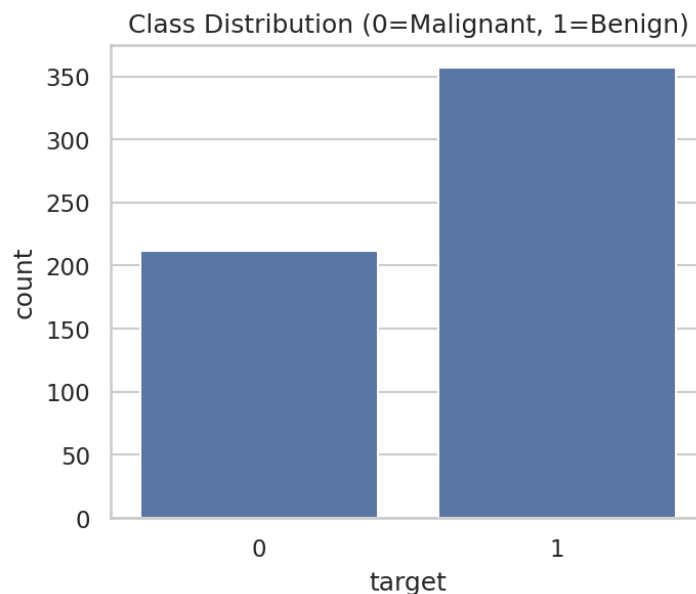


Figure 1: Class distribution of the target variable (0 = Malignant, 1 = Benign)

Inference: The dataset contains 357 benign and 212 malignant samples, showing a moderate class imbalance (roughly 63% vs 37%). This is not severe enough to require resampling, but stratified splitting and stratified cross-validation are used to preserve class proportions in every fold. The imbalance also motivates reporting precision, recall and F1-score in addition to accuracy.

Feature Correlation

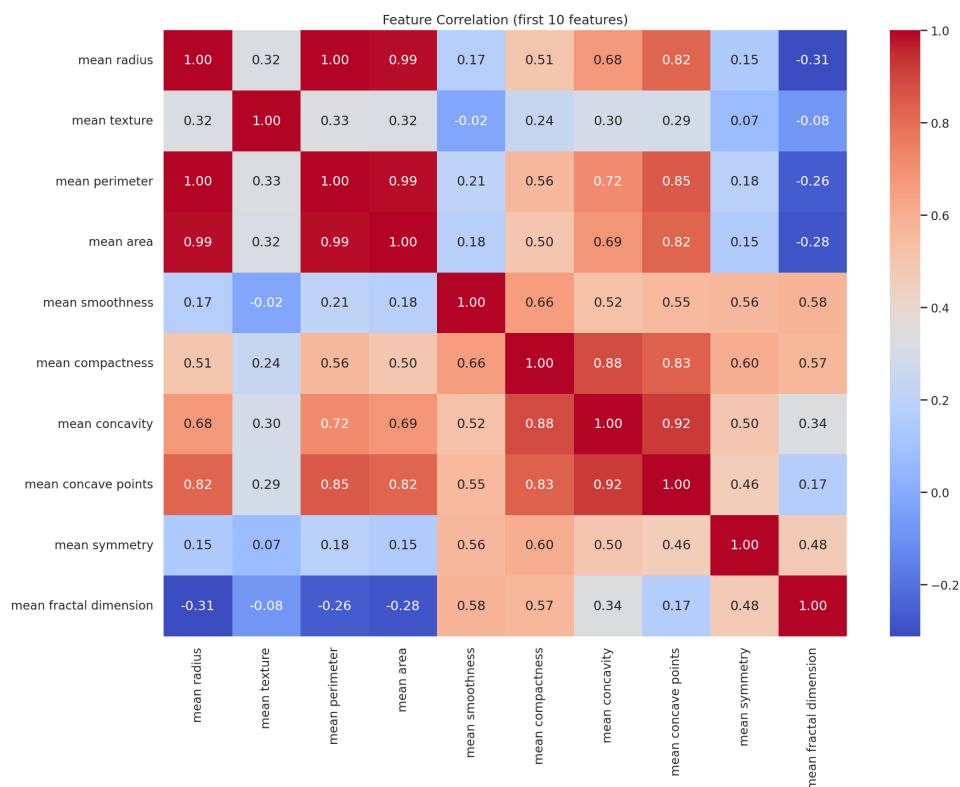


Figure 2: Correlation matrix of the first 10 features

Inference: Several size-related features such as **mean radius**, **mean perimeter** and **mean area** are very strongly correlated with each other (correlation close to 1), which is expected since they all measure cell size. Shape-related features such as **mean smoothness** show weaker correlation with the size features. This redundancy among features is well handled by tree-based models since they perform implicit feature selection at each split.

Feature Distribution

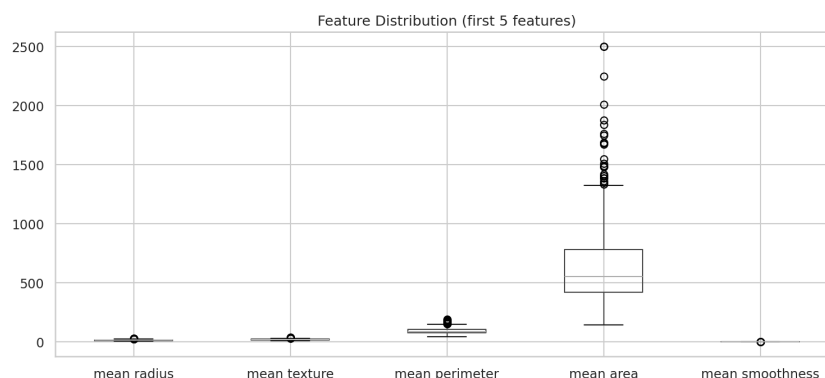


Figure 3: Boxplot distribution of the first 5 features

Inference: The features are on very different numerical scales (e.g. **mean area** is orders of magnitude larger than **mean smoothness**) and contain several outliers, visible as points

beyond the whiskers. Since Decision Trees and Random Forests split on individual feature thresholds, they are scale-invariant and do not require feature scaling, unlike distance-based algorithms.

5 Data Preprocessing

- **Missing value handling:** The dataset was checked and contains no missing values, so no imputation was required.
- **Encoding:** The target variable is already encoded numerically by scikit-learn (0 = Malignant, 1 = Benign); no additional label encoding was needed.
- **Feature scaling:** Not applied, since tree-based models are invariant to monotonic feature scaling.
- **Train-test split:** An 80–20 stratified split was used (`random_state=42`), giving 455 training and 114 test samples while preserving class proportions.
- **Feature engineering:** No additional feature engineering was performed; all 30 original numerical attributes were used directly.

6 Machine Learning Algorithm

Decision Tree Classifier

Working principle: A Decision Tree recursively partitions the feature space into regions with increasingly pure class labels. At each node, the feature and threshold that most reduce impurity are selected to split the data, until a stopping criterion (maximum depth, minimum samples per leaf, or pure nodes) is reached.

Mathematical formulation: For a node t with class proportions p_i , the Gini Index and Entropy impurity measures are defined as

$$\text{Gini}(t) = 1 - \sum_{i=1}^C p_i^2, \quad \text{Entropy}(t) = - \sum_{i=1}^C p_i \log_2 p_i \quad (1)$$

A split is chosen to maximize the information gain

$$\Delta I = I(t) - \sum_k \frac{N_k}{N} I(t_k) \quad (2)$$

where I is the impurity measure, N is the number of samples at the parent node, and N_k is the number of samples in child node t_k .

Advantages: Easy to interpret and visualize, handles non-linear relationships, requires no feature scaling, and can handle both numerical and categorical data.

Limitations: Prone to high variance and overfitting, especially when grown to full depth; small changes in data can produce very different trees; biased toward features with many levels.

Random Forest Classifier

Working principle: Random Forest is a bagging ensemble that builds many Decision Trees, each trained on a bootstrapped sample of the training data. At each split, only a random subset of features is considered, which decorrelates the individual trees. The final prediction is obtained via majority voting across all trees.

Mathematical formulation: Given B bootstrapped trees $h_1, \dots, h_B$, the ensemble prediction is

$$\hat{y} = \text{mode}\{h_1(x), h_2(x), \dots, h_B(x)\} \quad (3)$$

Because each tree is trained on an independent bootstrap sample and a random feature subset, the variance of the averaged predictor is reduced relative to a single tree:

$$\text{Var}(\bar{h}) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2 \quad (4)$$

where σ^2 is the variance of an individual tree and ρ is the pairwise correlation between trees, which random feature selection reduces.

Advantages: Reduces variance and overfitting compared to a single tree, provides feature importance estimates, robust to noise and outliers, generally higher accuracy.

Limitations: Less interpretable than a single tree, higher computational and memory cost, and slower prediction time for large numbers of trees.

7 Experimental Setup

Python Version	3.12
Scikit-Learn Version	1.x (as used in the Colab/Jupyter environment)
IDE	Jupyter Notebook
Operating System	Linux (cloud runtime)
Hardware	CPU-based execution

8 Hyperparameter Selection

Decision Tree Cross-Fold Results

Criterion	Max Depth	Avg CV Accuracy (%)	Avg CV F1 Score
gini	3	92.53	0.9400
gini	5	92.31	0.9373
gini	7	92.31	0.9375
gini	10	91.65	0.9319
gini	None	91.65	0.9319
entropy	3	93.41	0.9480
entropy	5	92.97	0.9439
entropy	7	92.97	0.9440
entropy	10	92.97	0.9440
entropy	None	92.97	0.9440

Best Decision Tree Parameters: criterion = entropy, max_depth = 3 (Avg CV Accuracy = 93.41%, Avg CV F1 Score = 0.9480).

Random Forest Cross-Fold Results

n_estimators	Max Depth	Max Features	Avg CV Accuracy (%)	Avg CV F1 Score
50	5	sqrt	95.60	0.9647
50	5	log2	95.60	0.9651
50	10	sqrt	95.60	0.9648
50	10	log2	96.04	0.9685
50	None	sqrt	95.60	0.9648
50	None	log2	96.04	0.9685
100	5	sqrt	96.04	0.9685
100	5	log2	95.82	0.9668
100	10	sqrt	96.26	0.9699
100	10	log2	96.48	0.9718
100	None	sqrt	96.26	0.9699
100	None	log2	96.48	0.9718
200	5	sqrt	95.60	0.9650
200	5	log2	95.82	0.9666
200	10	sqrt	96.26	0.9700
200	10	log2	96.04	0.9683
200	None	sqrt	96.26	0.9700
200	None	log2	96.04	0.9683

Best Random Forest Parameters: n_estimators = 100, max_depth = 10, max_features = log2, bootstrap = True (Avg CV Accuracy = 96.48%, Avg CV F1 Score = 0.9718).

9 Results

Baseline (Default Hyperparameters) Test Set Performance

Metric	Decision Tree (default)	Random Forest (default)
Accuracy	0.9123	0.9561
Precision	0.9559	0.9589
Recall	0.9028	0.9722
F1-score	0.9286	0.9655

Tuned Models – Test Set Performance

Metric	Value	
Accuracy	DT: 0.9474	RF: 0.9561
Precision	DT: 0.9459	RF: 0.9589
Recall	DT: 0.9722	RF: 0.9722
F1-score	DT: 0.9589	RF: 0.9655
ROC-AUC	DT: 0.9448	RF: 0.9924

10 Result Visualization

Confusion Matrices

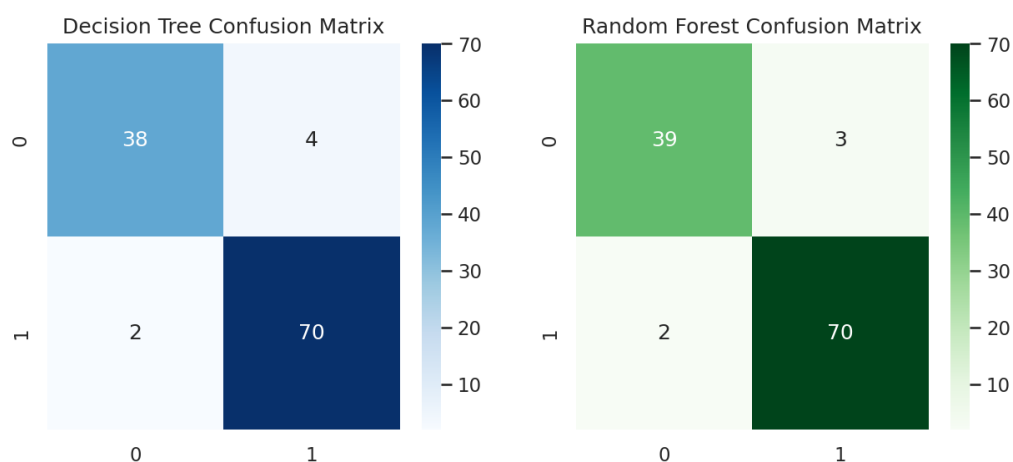


Figure 4: Confusion matrices for the tuned Decision Tree (left) and tuned Random Forest (right) on the test set

Inference: The Decision Tree misclassifies 6 samples (4 false positives, 2 false negatives) while the Random Forest misclassifies 5 samples (3 false positives, 2 false negatives). Both models achieve identical recall (missing the same 2 malignant cases), but the Random Forest produces fewer false positives, indicating slightly better overall discrimination between the two classes.

ROC Curve

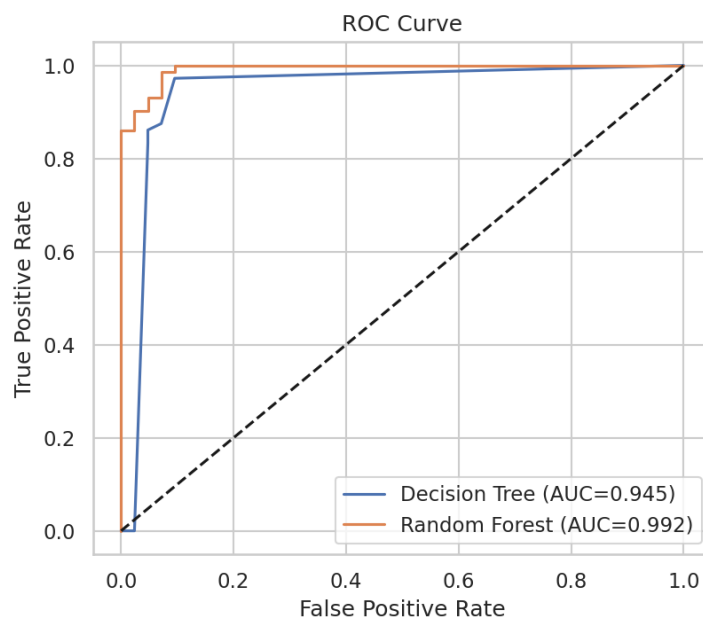


Figure 5: ROC curves for the tuned Decision Tree and Random Forest on the test set

Inference: The Random Forest achieves an AUC of 0.992, far closer to the ideal value of 1 than the Decision Tree’s AUC of 0.945. This shows that the Random Forest’s predicted probabilities separate the two classes much more reliably across all classification thresholds, since averaging over many trees produces smoother, better-calibrated probability estimates than a single tree.

Feature Importance

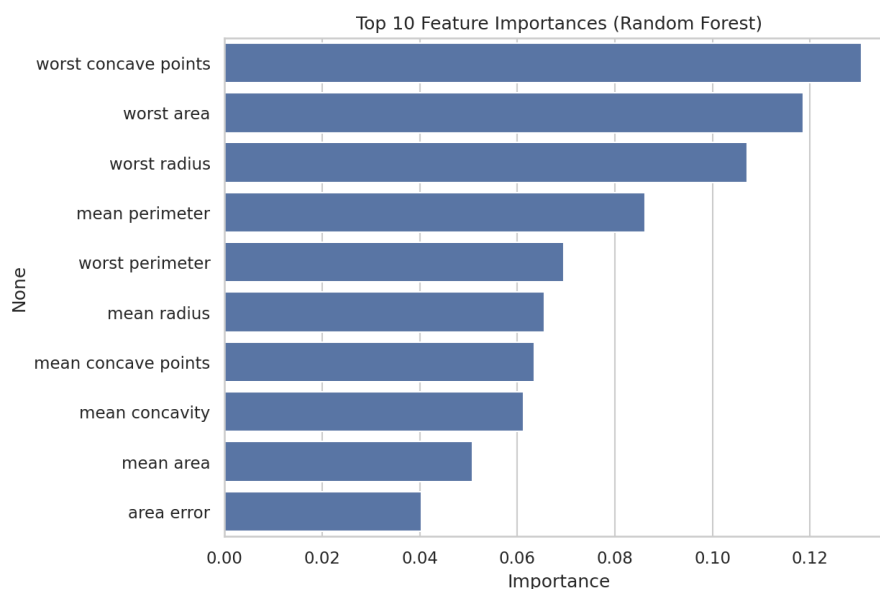


Figure 6: Top 10 feature importances from the tuned Random Forest

Inference: `worst concave points`, `worst area` and `worst radius` are the most influential features, consistent with clinical intuition that the severity of nuclear shape irregularity and overall tumor size are strong indicators of malignancy. These “worst” (largest observed) measurements dominate over mean and standard error measurements, suggesting extreme values in a sample are more diagnostic than average values.

5-Fold Cross-Validation Comparison

Fold	Decision Tree	Random Forest
1	0.9123	0.9649
2	0.8596	0.9561
3	0.9386	0.9561
4	0.9474	0.9561
5	0.9204	0.9646
Average	0.9156	0.9596

Inference: The Random Forest consistently outperforms the Decision Tree in every fold and shows much lower variance across folds (standard deviation is visibly smaller), directly demonstrating the variance-reduction benefit of bagging and random feature selection over a single decision tree.

11 Discussion

How does tree depth affect overfitting in Decision Trees? As max depth increases, the Decision Tree fits the training data more closely, reducing training error but increasing variance on unseen data. In the cross-validation results, accuracy and F1-score actually peak at a shallow depth of 3 and slightly decline or plateau at greater depths, showing that deeper trees begin to memorize noise in the training folds (overfitting), so a moderate depth generalizes best.

Which hyperparameter had the greatest impact on performance? For the Decision Tree, `criterion` and `max_depth` jointly had the largest effect, with `entropy` at `max_depth=3` giving the best F1-score. For the Random Forest, `max_depth` together with `max_features=log2` had the greatest impact, since restricting both tree depth and the number of candidate features per split reduced correlation between trees while avoiding overfitting.

How does Random Forest improve generalization? By averaging predictions over many trees trained on bootstrapped samples with random feature subsets, Random Forest reduces the correlation between individual trees' errors. This bagging mechanism lowers the variance of the ensemble prediction without substantially increasing bias, which is directly visible in the higher and more stable cross-validation accuracy (Section 8.4) and the higher ROC-AUC compared to the single Decision Tree.

Did ensemble learning always improve performance? Why or why not? In this experiment, the Random Forest outperformed the Decision Tree on every metric and every cross-validation fold, so ensembling helped consistently here. However, ensemble learning does not always improve performance: on very small or very simple, easily separable datasets a well-tuned single tree can match an ensemble, and the gain from bagging depends on how much variance (rather than bias) is limiting the base learner's accuracy.

12 Conclusion

Decision Tree and Random Forest models were implemented and evaluated using 5-fold cross-validation on the Wisconsin Diagnostic Breast Cancer dataset. Hyperparameters were selected based on average cross-validation F1-score, yielding a tuned Decision Tree (`criterion=entropy`, `max_depth=3`) and a tuned Random Forest (`n_estimators=100`, `max_depth=10`, `max_features=log2`). On the held-out test set, the Random Forest achieved higher accuracy (0.9561 vs 0.9474), higher F1-score (0.9655 vs 0.9589) and substantially higher ROC-AUC (0.9924 vs 0.9448) than the single Decision Tree, while also showing lower variance across the 5-fold cross-validation (std much smaller for RF). These results confirm that Random Forest's bagging and random feature selection reduce variance and improve generalization compared to a single Decision Tree, at the cost of reduced interpretability and higher computational complexity.

13 References

1. Scikit-learn Developers, "Decision Trees," *scikit-learn documentation*. [Online]. Available: <https://scikit-learn.org/stable/modules/tree.html>

2. Scikit-learn Developers, “Ensemble methods: Random Forests,” *scikit-learn documentation*. [Online]. Available: <https://scikit-learn.org/stable/modules/ensemble.html>
3. W. Wolberg, O. Mangasarian, N. Street, and W. Street, “Breast Cancer Wisconsin (Diagnostic) Data Set,” *UCI Machine Learning Repository*. [Online]. Available: <https://archive.ics.uci.edu>